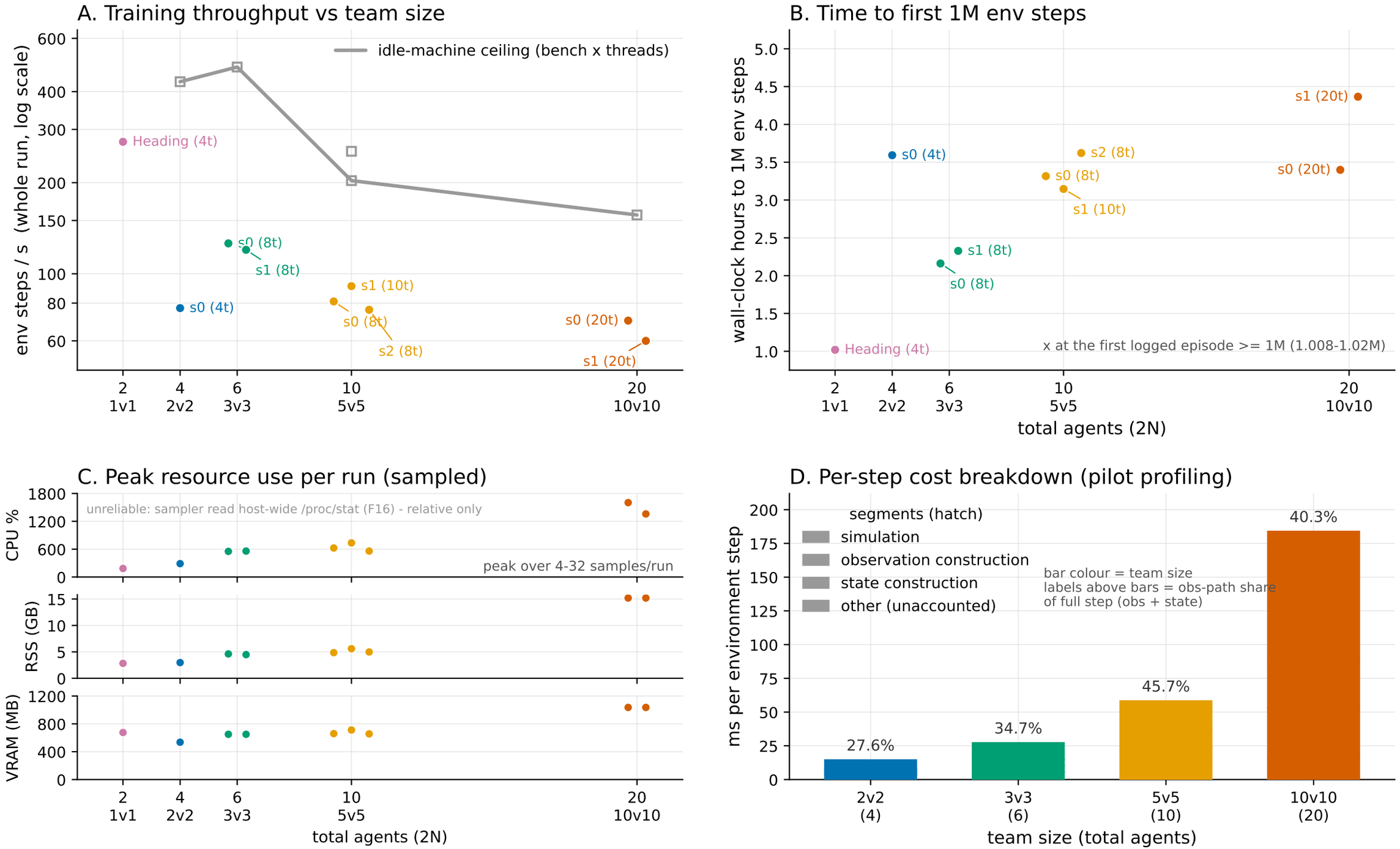


Panels A-C: the 8 scaling runs + the official single-agent heading control (N=1); the four official 1v1 combat runs (3 threads, a separate experiment family) are omitted. Rollout threads differ per run (4 / 8 / 8-10 / 20) and the campaign shared one 48-core box (~5 concurrent queues), so points are not an equal-compute comparison; point labels give seed and rollout threads. Three of the nine runs are partial (5.4M / 2.3M / 1.86M steps), the rest reached 6M. A: per-run median of the fps column (log scale); grey = pilot idle-machine single-env benchmark x that run's threads (a no-contention ceiling, not a training run; measured under 5-queue load the benchmark drops to 57-84% of idle). B: wall_time of the first episode at/above 1M steps minus the first logged row (undercounts by up to one episode). C: peaks over 4-32 samples per run; cpu_pct comes from the host-wide /proc/stat sampler (F16) - relative only, not container-accurate. D: independent pilot profiling, not sampled during training; 'other' is the unaccounted residual (full_step_ms minus sim + obs + state); share labels above bars = (get_obs + get_state)/full_step.